LLMLagBench: Identifying Temporal Training Boundaries in Large Language Models

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Abstract

Large Language Models (LLMs) are pretrained on textual data up to a specific temporal cutoff. This creates a strict knowledge boundary beyond which models cannot provide accurate information without querying external sources. More subtly, when this limitation is unknown or ignored, LLMs may inadvertently blend outdated time-sensitive information with general knowledge during reasoning tasks, potentially compromising response accuracy. We introduce LLMLagBench, an LLM freshness benchmark, as a systematic approach for identifying the earliest probable temporal boundaries of an LLM's training data by evaluating its knowledge of recent events. We then apply this benchmark to evaluate a large set of LLMs, including models with both explicitly declared and undeclared training cutoffs. The reliability of the benchmark is assessed by manual validation and comparison with publicly released information about LLM pretraining.

1 Introduction

LLM parameters are learned from naturally-occurring textual data, instructions and preferences. All of these sources of data contain temporally-bounded knowledge about the world. Although LLMs can be continually pretrained on texts and further finetuned on instructions, their adaptation can be both impractical (especially in the case of large base models) and potentially risky due to the problem of catastrophic forgetting whereby incremental addition of new data impairs previously acquired knowledge or abilities of a pretrained model [57]. As a result, major updates of base and instruct models are rare, and such models are effectively released with a strict knowledge cutoff date beyond which they are not updated until subsequent releases.[1]

The more obvious implications of this limitation, such as the tendency for LLMs to directly hallucinate about recent events, are usually addressed by providing retrieval-augmented answer generation mechanisms (RAG) [48]. However, there are potentially more subtle risks of interpolating outdated information with general knowledge in reasoning or forced-classification tasks. For example, a model can implicitly rely on recently outdated medical knowledge when making health-related recommendations. To illustrate, when Llama 3.1 70B-Instruct was indirectly prompted during our tests about whether blood lead levels of 15 µg/dL in manufacturing workers require intervention, it referenced outdated US OSHA standards without acknowledging that this value now represents the EU's biological limit threshold established in February 2024 occupational exposure regulations. The model provided guidance based on higher, less protective thresholds—all without indicating any uncertainty about its knowledge currency. This behavior illustrates how temporally-bounded knowledge can surreptitiously compromise decision-making in certain domains. Even models with access to search systems may either fail to recognize implicitly time-sensitive questions or retrieve incomplete results from search engines. It is therefore vital to know *the earliest probable training cutoff* of any LLM[2] used to generate knowledge-based answers and decisions.

In practice, determining an LLM's actual knowledge cutoff is not straightforward. While a model's release date provides a hard upper bound for open-weight models, it is less reliable for models accessible only through vendor APIs. Even when developers declare a single specific cutoff date, interpreting its practical implications remains challenging. Furthermore, instruction-following models can exhibit

[1]This problem is further explored in research on dynamically updating the knowledge base of LLMs [28] [50] [58].

[2]Our benchmark indicates October 2023 as a likely cutoff point for both LLama 3.1 and 3.3.

contradictory behavior: they may declare overly conservative cutoff dates while simultaneously hallucinating about events far beyond their actual training period.[3] The situation is further complicated by the fact that knowledge infusion may operate differently during pretraining, continued pretraining and post-training phases. Our analysis reveals that several LLMs exhibit multiple partial cutoff points, possibly corresponding to these distinct training stages. These observations motivate the need for a systematic empirical approach to identifying LLM knowledge boundaries as the central contribution of this work.

2 Previous work

A number of existing LLM evaluation benchmarks assess time-sensitive knowledge [38, 24], temporal reasoning capibilities [51, 52, 11] and awareness [60].

Recent work by Zhu et al. [59] introduced the concept of temporal generalization in LLMs, examining how models perform on data before and after their release dates. While their approach evaluates temporal bias and generalization by assuming the cutoff date equals the model release date (which is in fact the latest possible cutoff), the work reported in this paper addresses a more fundamental question: empirically identifying when a model's training data actually ends. Our analysis reveals that training cutoffs often diverge significantly from release dates and that models frequently exhibit multiple partial cutoffs corresponding to different training phases.

Cheng et al. [10] introduced perplexity-based probing to operationalize and identify "effective cutoffs" in LLMs, revealing significant temporal misalignment due to deduplication failures and CommonCrawl contamination with outdated content. Their approach requires white-box access to model probabilities, limiting evaluation to open-source models.

CutoffBench (`https://cutoffbench.com`) is a publicly available benchmark which provides an automated approach to detecting LLM knowledge cutoffs using sports and financial time-series data with binary scoring and threshold detection. As explained further, in contrast to this benchmark we use PELT changepoint detection to enable identification of multiple partial cutoffs rather than assuming a single transition point. This is potentially important because many LLMs exhibit knowledge boundaries corresponding to different training phases. Second, our manually curated questions span diverse news domains rather than specialized areas like sports and finance, providing broader coverage of general knowledge. We also explicitly track refusal behavior, distinguishing instruction-tuned refusal from genuine knowledge absence.

3 *LLMLagBench*

This paper presents LLMLagBench, a benchmark designed to systematically identify the temporal boundaries of an LLM's training data through systematic evaluation of its knowledge of recent events. Using a set of manually selected questions densely sampled from news reports published over the last five years we calculate significant performance changepoints in answer accuracy.

3.1 Data Collection and Filtering

As of October 2025, the benchmark comprises 1,713 questions about events that could not be accurately answered before they were reported in news media. Representative examples of such question-answer pairs include:

- (January 8, 2024) On what date did Adan Canto pass away? $\rightarrow$ Adan Canto passed away on January 8, 2024.
- (December 4, 2024) Who was fatally shot outside a New York City hotel early on December 4, 2024? $\rightarrow$ UnitedHealthcare CEO Brian Thompson was fatally shot outside a New York City hotel early on December 4, 2024.
- (April 3, 2024) What did John Legend perform with Ukrainian singer Mika Newton and poet Lyuba Yakimchuk during the 2022 Grammy Awards? $\rightarrow$ John Legend performed his song 'Free' with Ukrainian singer Mika Newton and poet Lyuba Yakimchuk during the 2022 Grammy Awards.

Although some of these questions could be guessed with some probability without access to later knowledge (including the one about the performance of Ukrainian artists), we considered the exact answers to these questions to be sufficiently unpredictable.

We sampled a total of $\simeq$ 80,000 news items from the period 2021-2025 and clustered them by topic

[3] Table 3 juxtaposes release dates, provider and model declarations.

within each day. On average each selected story was reported by 10 distinct news outlets (including at least one from our subset of worldwide/reliable sources). We then used **deepseek-ai/DeepSeek-V3-0324** to extract approx. 8,400 candidate questions from these articles. Table 1 summarizes the distribution of news outlets represented in the question set.

Despite detailed prompting, the LLM-based extractor occasionally generated questions that were trivial to answer, typically referring to regularly scheduled events or previously reported outcomes, as illustrated below:

- (May 5, 2021) On what date in 2021 did Cinco de Mayo occur?
- (April 9, 2022) Which horse won the 2021 Grand National at Aintree?

To ensure benchmark quality, we manually validated all candidate questions and selected 1,713 question-answer pairs that met our criteria for temporal specificity and unpredictability. Table 4 presents additional examples illustrating the distinction between questions that meet our unpredictability criteria and those that were rejected as easily answerable.

3.2 Answer Evaluation

Each LLM evaluated in LLMLagBench is prompted to answer all 1,713 questions (approx. 7 questions per week) using the following standardized prompt:

> *Produce a concise answer to the following question. Generate only the answer, no additional comments. Don't speculate. If you don't know the answer, simply write 'I don't know'.*
>
> *Question: [QUESTION]*

We then use **deepseek-ai/DeepSeek-V3-0324** as an evaluator to assess these answers. The evaluation prompt instructs the LLM to rate each answer across three dimensions:

- **Factual Accuracy** – Are the facts in the answer correct? (0 = completely wrong or contains major factual errors; 1 = partially correct; 2 = fully accurate)
- **Relevance** – Is all content related to the question? (0–2 scale with rating guidelines)
- **Faithfulness to the Gold Answer** – Does the answer align with the reference answer? (0 = no overlap with the ideal answer; 1 = partial overlap, capturing some but not all core ideas; 2 = fully aligned with all core ideas)

Of these three criteria, LLMLagBench currently uses only **Faithfulness to the Gold Answer** as the primary evaluation metric. In a separate iteration, however, the evaluator model identifies instances where the evaluated models produce **refusals to answer** or otherwise evasive responses.

Source	Candidates	Selected
bbc.com	325	58
cnn.com	200	43
dailymail.co.uk	1529	277
foxnews.com	151	41
independent.co.uk	2862	594
nypost.com	839	169
nytimes.com	454	81
theguardian.com	783	187
usatoday.com	456	98
washingtonpost.com	301	62
washingtontimes.com	515	103
Total	**8,415**	**1,713**

Table 1: Question sources and selection statistics.

To assess the reliability of the automated evaluation, we measured inter-rater agreement between the evaluator model's faithfulness ratings and those provided by two human annotators on a subset of 500 questions across different versions of the evaluation prompts. The final configuration achieved Cohen's Kappa scores of 0.81 and 0.83, respectively. The generation temperature of the evaluator model was tuned to maximize these inter-rater agreement values.

3.3 Training Cutoff Detection

Using the optimized generation hyperparameters of the evaluator model, we calculated faithfulness scores ranging from 0 to 2 for all evaluated models. While these scores can be averaged over specific time periods to estimate a model's performance on time-sensitive knowledge questions, the primary objective of LLMLagBench is to provide an independent estimate of the earliest probable training cutoff point.

Our approach is based on the assumption that models should exhibit a sudden, significant decrease in performance beyond their training data